

The Mass of Matter Gets a Receipt

The knot solver, the conditioning dictionary, and the two-term mass model. Claims FND-MATTER-006 through FND-MATTER-014; benchmarks in benchmarks/foundations/. Every number in this paper is either an exact theorem, a certified constant, or a measurement with its instrument and its failures named.

1. The question, and the fork it became

This programme was asked, in plain words, whether it could predict the atomic weight of hydrogen. The honest answer was no: absolute masses require the absolute strand scale, blocked at FND-MATTER-003. But hydrogen is 99.95 percent proton, so the question sharpens into a dimensionless one -- the proton-to-electron mass ratio, 1836.15 -- and dimensionless ratios are exactly what knot geometry can chase without any scale at all. That reframing, plus a first arithmetic confrontation, produced the fork that organizes this paper (FND-MATTER-007): if a particle's rest energy were purely tension times rope length, a trefoil electron would demand a proton of ropelength roughly thirty-one thousand -- a thousand-crossing-class tangle -- while a minimal-ring electron would demand roughly six thousand. Either the proton is an enormous tangle, or mass has structural contributions beyond length. Both branches are honest physics; the corpus owns the fork explicitly, and the remainder of this paper builds the instruments that make the second branch computable.

2. The knot solver, and a spectrum with its failures kept

The gateway instrument (FND-MATTER-006): a SONO-class tightening solver -- edge equalization, a curvature cap enforcing turning radius at least half the rope diameter (the local half of thickness), uniform shrink, and non-local excluded volume applied only beyond the $\pi D/2$ arc boundary of legitimate near-contact, with contact-averaged damped overlap resolution. Validated the house way, control first: the unknot closes at the geometric bound π to 0.04 percent, which is what earned the trefoil a hearing. First light: the tight trefoil at $L/D = 16.84$ against the ideal-knot literature's 16.372 -- 2.9 percent, inside the locked bar, topology provably preserved throughout.

The first spectrum session (FND-MATTER-007) is registered as what it was: two misses and an anomaly. The torus knot 5_1 lands at 25.09 against roughly 23.55 -- 6.5 percent, MISSING its locked five-percent bar, kept as a miss with the bar unwidened. And the figure-eight, the table's first non-torus knot, robustly converges to 31.9 from the standard embedding -- annealing-invariant to a tenth of a percent, no pass-throughs, cause unresolved. The stall value is asserted in the benchmark so that any future solver change that silently moves it is flagged:

kept negatives are tracked, not forgotten. The first spectroscopy ratio stands: $E(\text{trefoil})/E(\text{ring}) = 5.36$ against the ideal 5.21, a pure geometry number requiring no scale.

3. The zero-point instrument, and the 43 percent verdict

A knot does not merely occupy the medium; it conditions it, and a conditioned region shifts every wave mode of the weave. Half the sum of those shifts is a Casimir-class self-energy the knot carries as surely as its arc length -- and on a lattice that sum is EXACT, the band edge serving as the regulator nature provides (FND-MATTER-008; stable to $3e-7$ under doubling). The measured structure: a bulk term per unit length plus a negative boundary correction, and -- the verdict -- equal-length, equal-total-strength defects of different SHAPE differ by 43 percent in their zero-point term, because the band saturates (four times the strength buys less than three times the shift) and concentration therefore matters. Any mass with a material zero-point component refuses to be a length: two knots of equal rope can weigh differently by their tightness alone. The thousand-crossing arithmetic of Section 1 is thereby NOT binding.

4. The two-term mass model

The marriage of the two instruments (FND-MATTER-009): $m(K) = \text{Sigma } L(K) + \text{lambda } dE_{zp}(K)$, with the conditioning profile extracted from the actual tightened geometry and lambda the relative weight that remains honestly blocked. The first table -- ring, trefoil, 5_1 -- delivers the lever: zero-point fractions DIFFER between knots by 25 percent (the tighter trefoil packs more turning per unit rope, and the vacuum notices), so mass ratios are lambda-dependent, running from 1.49 at pure length to 1.16 at zero-point dominance. Geometry alone does not fix the mass spectrum -- and the ratios COMPRESS toward one as the zero-point weight grows, which is qualitatively the shape of the actual baryon table, noted as a direction and not a fit.

5. The conditioning dictionary: two guesses become two theorems

5.1 The contact term, exactly

A contact couples two strands through their RELATIVE displacement and ignores their sum -- so in parity variables the symmetric sector passes through a crossing untouched (verified identically zero off-diagonal; symmetric spectrum equal to the free chain at $5e-14$) while the antisymmetric sector sees pure on-site pinning at twice the contact stiffness. Hence, exactly: a crossing of stiffness k_c over w sites equals a single-chain pinning defect of strength $2 k_c$ over w sites (FND-MATTER-010, verified at $1e-13$). Every contact in every knot conditions identically, with k_c one material constant for the entire spectrum. Freedom removed from the model is precision added to every future ratio.

5.2 The bend term: a refutation, a conviction, and a closed law

The curvature term was written down as κ^2 -- and the engine refuted it (FND-MATTER-011): on a boundary-free instrument (a pre-tensioned circle against a periodic straight chain, dead loads outside the Hessian), the quadratic coefficient drifts 66 percent across sizes. The raw signal instead tracks one over root L across a fourfold range -- and that apparent fractional exponent was then CONVICTED as a logarithm (FND-MATTER-012): the circle's Hessian block-diagonalizes exactly into 2×2 blocks per angular index, the per-mode ledger closes on the total at six decimals, and the law $dE \times L = a + b \ln L$ fits every size at two parts in ten thousand while predicting held-out sizes at the same precision. The mechanism, read off the ledger, is a competition: the circle stiffens its lowest transverse modes (the block determinant carries the exact $(\cos q - \cos \alpha)^2$ invariant, vanishing at the translations) while softening the high band by a small per-mode drift, and the logarithm is born of the contest.

Both constants are now derived. The slope (FND-MATTER-013): $b = 2 \pi (5/2 - 7 \sqrt{2})/4 = 0.157866$, obtained symbolically as the pole of the α^2 -squared kernel at fixed Bloch phase, with five determinations in a line -- analytic blocks validated at $9e-16$ in eigenvalues, a decade-wide plateau, the exact symbolic limit, the block sums, and the original lattice measurement. The intercept (FND-MATTER-014): $a = -0.509658$, a certified constant with an exact convergent representation -- thirty-digit low-mode blocks spliced onto the perturbative kernel at a point where both agree to six digits, converging on a clean $1/N$ ladder to $1e-11$. The closed-form question is answered honestly: three different constant bases return three contradictory candidates, all rejected as lattice-reduction artifacts; a joins the family of lattice Casimir constants. THE BEND LAW, FINAL, ZERO FREE PARAMETERS: $dE \times L = -0.509658 + 2 \pi (5/2 - 7 \sqrt{2})/4 \ln L$.

6. Honest limits, instrument lessons, and the road

What is NOT established, at full volume: no knot is identified with any particle -- the assignment is the era's central question, with named hazards (composite knots, the two-strand framing, the figure-eight anomaly). The relative weight λ and the contact stiffness k_c are the campaign's two remaining unknowns -- both material properties, neither per-knot freedom -- with λ the blocked absolute scale wearing its matter-sector costume, and now visibly the last conceptual gate between this corpus and the mass of hydrogen. The bend analysis is $1+1$ and at the canonical rope point; the general (k, T) forms are named next-orders.

Five instrument lessons were paid for and filed (catches eighteen through twenty-two): the local/non-local distinction every thickness code must make; boundary contamination at small signals, cured by closed loops; the rotation that is not a symmetry (a ring held by external contacts cannot rotate for free --

a knot's bent segments are stiffer than symmetry-counting suggests); unitarily equivalent blocks share eigenvalues, not entries; and the intercept lesson -- instruments that agree on logarithmic slopes may still disagree on constants, so certify each separately. The road, in registry order: convict the figure-eight; the near-degenerate pair rehearsal for the neutron-proton split (1.00138 -- exactly the kind of third-decimal splitting a shape-sensitive mass term produces naturally); kc from the three-dimensional pressing geometry; the general-parameter dictionary; and the scale -- blocked, and said so on every claim it touches.

7. The instrumentation epic: certificates, constructors, and movers

Between v2.3.2 and v2.3.3 the campaign built its identification, construction, and rearrangement machinery, judging every tool against ground truth. The ALEXANDER CERTIFIER evaluates the Wirtinger crossing relations at $t = 2$ in exact integers; because Alexander polynomials are defined up to units and t to $1/t$, the odd part of the absolute value is the convention-proof statistic, and the validation table showed every raw value differing between projections by a pure power of two -- the theory's unit ambiguity appearing in the data exactly as predicted. The famous vanishing of the 6_1 polynomial at $t = 2$ turned the determinant test's pathology into a fingerprint, and the two-certificate pipeline (determinant plus Alexander) made identity rigorous where determinants collide: 7_2 from 6_2 (both det 11; odd parts 5 and 1), 8_1 from 6_3 , the triple composite returning $27 = 3$ cubed unprompted -- the polynomial's multiplicativity under connected sum, a theorem the instrument was never taught, materializing as an integer.

The PLAT CONSTRUCTOR opened the two-bridge family (including braid-index-4 knots the trace closure cannot reach), and a 48-word certificate scan returned the entire twist ladder with its anatomy visible -- one clasp prefix plus a growing twist block. Its negative was kept at full volume: plat basins are uniformly shallower than trace-closure basins; topological nativeness does not confer basin quality; descent inherits geometry, not meaning.

The mover campaign ran three claims: certifier-gated crankshaft basin hopping (safe through thousands of pi-scale rotations -- zero topology corruption -- and insufficient, its shortfall localizing the failure in the move class); discrete transport (safe, shelf-bound); and REPTATION -- the conveyor anneal, a coherent tangential drift inside the constraint loop, specified by its predecessors' kept failures and confirmed on an answer key to one hundredth (the plat figure-eight carried to 21.543 against the deep floor's 21.533). Deep rearrangements in tight knots are material transactions, not spatial swings. The flow-amplitude threshold proved to be a physical measurement, dropping with clasp congestion, and grew into the CONGESTION GAUGE: a ladder of flow-stability thresholds (0.12, 0.05, 0.03) ranking knots by clasp congestion using nothing but the corpus's own dynamics -- a literature-free characterization of

knot geometry. The 6_1 and 7_2 walls survived eleven distinct attacks across two eras and a resolution ladder, and were terminal-graded as measured solver-landscape features; surpassing them is solver engineering, not physics.

8. The zero, the library, and the corrected table

The resolution-scaling test turned a nuisance into an instrument: the figure-eight ladder (21.58, 21.53, 21.52 at $N = 130, 140, 200$) converges to a resolution-stable +2.3 percent solver systematic -- five external anchors logged at 2.5 to 2.9 percent were ONE property reproduced five times. Calibrated out, the corrected table's most striking cell is a zero: the figure-eight at 21.039 against the literature's 21.040, the fourth significant figure, with five stacked subsystems all required simultaneously sound for it to emerge. Thirteen knots stand certified at the table, 8_1 seated last.

The reference column was then upgraded from recall-grade to source-grade (Cantarella, LaPointe, and Rawdon's ridgerunner computations; units verified on the tight trefoil to four figures), and the library returned a fact nobody ordered: THE IDEAL DOUBLET IS NEARLY DEGENERATE -- the granny at 28.520 diameters and the square at 28.535, split by five hundredths of a percent, the granny marginally shorter. For a program that had cast exactly this pair as its isospin rehearsal, a striking datum -- and a coincidence of configuration until a mechanism connects tight-knot geometry to mass, which the paper states as plainly as the registry does. The conditional built on it faced two pre-barred juries (universality, sign, an unrehearsed rescue of a drowning mass defect, junction spread failed at full volume) and survived on its observables.

9. The gate saga: the axis becomes a number

The scale λ -- blocked since the model's first table -- was then taken through the longest single arc in the corpus. THE AUDIT first: the bend and contact constants descend from the same minus-chain lattice with the same normalization, so the conditioning ledger has no internal freedom and the gate is exactly one number. THE COLLAPSE: recognizing the mode lattice as a beaded string under the rope's own tension ($kt = T/a$) and the framework's torque-waves-at-c postulate as the relativistic-string closure ($\mu = T a \text{ over } c \text{ squared}$) reduces λ to $\hbar c \text{ over } (a T D)$. THE KILL: the collapsed gate's first act was to audit the doublet hypothesis, and under the quantum reading it forces a mesh scale near 200 femtometers against the framework's own Lorentz bound of a tenth of a femtometer -- a violation by three orders of magnitude, robust across every granted ambiguity. The kill is internal, arithmetic, and was registered the day after the coincidence was banked.

THE FORK was then worked out by exact computation. The surviving classical-amplitude reading carries a TRACE ledger whose contact term is exactly minus two per frozen site at every patch width -- perfectly extensive where the killed

reading was patch-nonlinear -- while its bend ledger vanishes identically (the trace is twice the sum of couplings, unmoved by magnitude-preserving tilts). The bend logarithm, the era's first derived beauty, is eulogized: spectrum truth, energetically silent, available for resurrection. Rerun end to end, the doublet condition proved nearly reading-blind (a three percent shift at matched seats against a two hundred percent basin scatter): the audit killed a reading without wounding the phenomena.

The remaining parameter -- the jitter amplitude as a fraction of tube diameter -- was then confiscated by COLLISION SATURATION: amplitudes are frequency-blind because geometry does not care about frequency, and fixed because the gap is the gap; the Monte Carlo shows a genuine thermal-to-saturated crossover with the saturated value drive-independent. The confining mesh was located by elimination from the framework's own ontology: ambient single ropes cannot confine (interpenetrable by prior commitment), and a new instrument-grade measurement showed tight tangles in contact essentially everywhere (nearest opposing strand at 0.998 diameters at every site, all knots) -- no jitter room in self-packing -- leaving only the tube's INTERNAL BUNDLE, where the density-onset principle closes the parameter exactly: the internal packing ratio equals the square root of the coverage threshold, and $\Lambda = (1 - \sqrt{f_c})^2$ over 12.

THE DETERMINATION, finally, was a direct percolation simulation of the transparency-loss principle in the framework's own geometry: constituent ropes as a Poisson field of width- w disks, a transverse test rope passing if its center finds a channel, tangibility onset the number density at which crossing paths vanish. The measured threshold (grid-corrected against the literature disk-percolation filling 1.128) gives $\Lambda = 0.013$ to 0.017 with ZERO free parameters, against the phenomenological window 0.014 to 0.044. The class verdict: percolation -- the window lives. The quantitative verdict: an edge landing that brackets the doublet condition at one seat basis and undershoots at another -- too close for dismissal, too edged for celebration, and squarely a seat-quality question. The mass model's conditioning sector is now parameter-free: $m = T D (L - \Lambda \ell)$, with Λ computed and $T D$ the model's single input.

10. The state of the sector, and the road

What is derived: the solver systematic; the beaded-string and wave-speed identities; the quantum-reading kill; the trace ledger's exactness; the saturation formula; the closure $x = \sqrt{f_c}$; the percolation threshold. What is measured: the walls, the congestion gauge, the self-packing of tight tangles, the edge landing. What remains hypothesis-shaped and is said so on every claim it touches: doublet $= n/p$; nucleon = tangle of 27.75 diameters; the medium above the saturation crossover; reading C itself. What remains open: the single scale input $T D$; the three-dimensional wiggle correction to the threshold; the constituent-count budget against the atomic N ; and the production-grade solver

that would turn edge landings into verdicts and junction noise into junction physics. No knot is identified with any particle. The road runs through better seats -- and through one number, T D, which now carries the entire absolute scale alone.